TOSHKENT SHAHRIDAGI INHA UNIVERSITETI
INHA UNIVERSITY IN TASHKENT

–1 POINTS FOR EACH FIELD LEFT EMPTY OR FILLED WRONG

ID NUMBER:

FIRST NAME: LAST NAME:

STUDENT SIGNATURE: ROOM NUMBER

FINAL EXAMINATION

COURSE NAME:	DISCRETE MATHEMATICS
COURSE NUMBER:	MSC 2050
EXAMINATION DURATION:	1 HOUR
ADDITIONAL MATERIALS ALLOWED TO BE USED:	–
SPECIAL INSTRUCTIONS	(1) FOLLOW ANY INSTRUCTIONS GIVEN BY PROCTORS. (2) ONCE YOU ARE ALLOWED TO OPEN THE EXAMINATION BOOKLET, YOU MUST PUT YOUR $\boxed{\text{ID}}$ ON EVERY ODD PAGE (3) READ THE INSTRUCTION CAREFULLY! Sometimes it is asked to provide JUST an answer, sometimes **FULL** solution.

- Final answers must be written by **only blue or black, non-erasable pen**.
- Do NOT use highlighters or correction pen.
- IUT's Examination Rules and Regulations apply.

Do NOT open the examination paper until directed to do so!

1. (8 points) For each statement below decide if it is TRUE or FALSE. Write **T** or **F** in the box at the right from the corresponding statement. Each correct answer is of 1 point.

   Given a conditional statement $p \to q$. The inverse of its contrapositive is $\neg q \to \neg p$

   $\forall x(x \neq 0 \to x|100)$ over the domain of positive integers.

   $\exists x(x$ is prime $\wedge\, x|100)$ over the domain of positive integers.

   If $A = \{1, 2, 3, 4\}$ and $B = \{4, 5, 6\}$ then the cardinality of the union $A \cup B$ is 7.

   It is possible to construct a one-to-one function from $\{1, 2, 3, 4\}$ to $\{a, b, c, d\}$.

   If A, B and C are sets, then $(A - B) - C = (A - C) - B.$

   The functions $n^2 + 2^n$ and $n^2 + 2^{100}$ are of the same order (have the same big-O estimate).

   One of solutions of the linear congruence $7x \equiv 13(mod 19)$ is $x = 10$.

2. (8 points) About the Triangular numbers T_n we are given that $T_1 = 1$ and also

$$T_n = T_{n-1} + n, \quad n > 1.$$

(a) (3 points) Check the formula

$$T_n = \frac{n(n+1)}{2}$$

for $n = 1$, $n = 2$ and $n = 3$.

(b) (5 points) Using the Principle of Mathematical Induction prove the formula

$$T_n = \frac{n(n+1)}{2}$$

for all $n \geq 1$.

3. (8 points) The sequence $\{a(n)\}$ is defined recursively as follows: $a(0) = 1$, $a(1) = 2$ and

$$a(n) = 2a(n-1) - a(n-2), \quad n \geq 2.$$

(a) (4 points) Write a recursive algorithm in pseudocode for finding $a(n)$.

(b) (4 points) Find an explicit formula for $a(n)$.

4. (8 points)

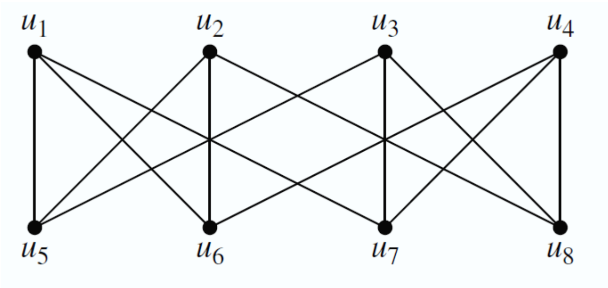
   (a) (1 points) Define relation on a set A.

   (b) (2 points) Define an equivalence relation.

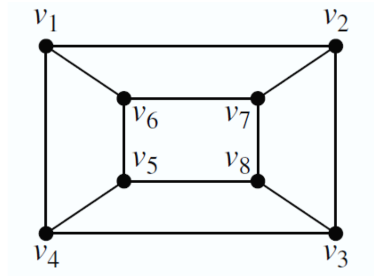
   (c) (2 points) Define a partial ordering.

   (d) (3 points) Find the reflexive, symmetric and transitive closure of the relation $\{(1,2),(2,3),(2,4),(3,1)\}$ on the set $\{1,2,3,4\}$.

5. (8 points) Consider two graphs:



G



H

(a) (4 point) Show that G and H are isomorphic.

If needed you may use result of (a) to answer the following questions:

(b) (1 point) Is G planar?

(c) (1 point) Is H bipartite?

(d) (1 points) What is chromatic number of H?

(e) (1 points) If I join subsequently $v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8$ will it be a spanning tree of H?

Do NOT fill	P.1	P.2	P.3	P.4	P.5	Total
max points	8	8	8	8	8	40
your score						